

Product Requirements Document (PRD): ALTA AI Unified Hospitality Platform

1. Executive Vision and Market Strategic Context

The Saudi Arabian hospitality sector is currently witnessing a tectonic shift, catalyzed by the mandates of Vision 2030. With the kingdom's hotel room inventory projected to reach 167,500 by the end of 2025, and a staggering 99,500+ additional rooms currently under construction or in planning stages, the scale of growth is unprecedented. Crucially, as an architect of this digital transformation, we must recognize that this new supply is heavily concentrated in the **Luxury and Upper-Upscale segments** (Source Image 2). High-end service at this volume cannot be sustained by human capital alone; it requires a sophisticated "Operating Brain" capable of scaling intimacy and precision. ALTA AI is engineered to bridge the widening "Market Gap" where global legacy systems fail to address the specificities of the Saudi market, particularly during the high-pressure surges of Haj and Umrah seasons. | Strategic Dimension | Traditional Fragmented Systems (Oracle, Mews, etc.) | ALTA Unified Brain Approach || ----- | ----- | ----- || **Operational Scope** | Strong on bookings, but **weak on post-departure management** and experience (Source Image 47). | Full-journey lifecycle management, from pre-arrival to automated post-stay recovery. || **Linguistic Depth** | Basic translation; fails to grasp nuances of Saudi dialects or voice-based intent. | Deep NLP for local dialects; native support for voice notes and sentiment detection. || **User Friction** | Requires complex staff training and proprietary guest app downloads. | "No-App" strategy; 100% WhatsApp-native interface for guests and staff. || **Data Utility** | Passive silos; data is "silent" and requires manual extraction for insights. | Proactive intelligence; an integrated dashboard that converts data into executive recommendations. |

This alignment is a market necessity. ALTA transforms the operational strain of religious tourism into a streamlined, automated workflow, moving the industry away from "Operational Chaos" and toward high-margin efficiency.

2. Problem Statement: The Operational Crisis

Modern hoteliers are trapped in a "Technological Illusion." While properties are often "over-tooled," they remain under-optimized. The presence of multiple disjointed software licenses creates **"Micro Friction"** (Source Image 31)—a state where staff spend more time navigating interfaces than serving guests. This fragmentation results in a "torn experience" where guest preferences are lost between the PMS, CRM, and task management tools.

Identified Points of Operational Friction:

- **Data Fragmentation:** Departments operate in isolation, leading to inconsistent service delivery and lost revenue opportunities.
- **IT Resource Scarcity:** The "Forgotten Middle" (Independent and Medium hotels with 50-250 rooms) possesses the budget for tech but lacks the massive IT teams required to maintain enterprise global stacks (Source Image 21).
- **Training Attrition:** High turnover in hospitality makes the constant training of complex, non-intuitive systems a significant financial drain.

- **Delayed Response:** A disjointed flow results in slow guest response times and a total lack of follow-up after the guest departs, damaging the property's digital reputation. These hotels do not need more software licenses; they need a **"Digital Workforce"** —an autonomous layer that executes tasks without human intervention.

3. Product Philosophy: The Digital Workforce Concept

The ALTA philosophy represents a fundamental pivot in hospitality tech: we are not selling a tool; we are "hiring out" a full team of AI agents. ALTA functions as a **24/7 Smart Operating Brain** that never sleeps, never forgets, and continuously optimizes the property's performance. **Core Pillars of the Digital Workforce:**

1. **Constant Availability:** 24/7/365 coverage of all guest touchpoints, ensuring zero missed inquiries even during peak Hajj/Umrah occupancy.
2. **Deep Localized Intelligence:** An inherent understanding of Saudi hospitality norms and linguistic context.
3. **Unified Command:** Connecting disparate departments into a single, cohesive intelligence layer. **The "Non-Replacement" Mandate:** Architecturally, ALTA is a non-disruptive "Intelligence Layer." A rigid technical requirement of the platform is **Integration over Replacement** (Source Image 9). ALTA sits above existing Property Management Systems (Oracle, Mews, etc.), gathering data via API to provide a sophisticated brain without the high-risk friction of a full system migration.

4. Functional Requirements: The AI Agent Hierarchy

ALTA's digital workforce operates under a collaborative hierarchy where agents share context under the supervision of a central executive. | Agent Role | Responsibilities | Value Proposition ("So What?") || ----- | ----- | ----- || **Executive Manager (AI-Modir)** | Central dashboard; daily reports on occupancy and revenue; proactive operational recommendations. | Eliminates manual data digging; identifies ROI opportunities before the shift begins. || **Reception Agent** | Inquiry handling; booking management; automated check-in/out follow-ups. | **Reduces reception calls and physical congestion** (Source Image 10); ensures 100% lead capture. || **Guest Service Agent** | Processes requests (cleaning, maintenance, late check-out); handles complaints via WhatsApp. | Drastically speeds up response times; prevents minor issues from escalating into negative reviews. || **Digital Reputation Agent** (وكيل السمعة الرقمية) | Monitoring Google Reviews; automated smart responses; reputation recovery. | Directly drives higher rankings and visibility, which correlates to higher ADR and RevPAR. || **Marketing & Sales Agent** | Content creation; campaign analysis; direct booking recovery and upselling. | Recovers "lost" guests and maximizes guest lifetime value through automated, personalized offers. || **Maintenance & Housekeeping Agent** | Task distribution; fault analysis; monitoring service delays. | Optimizes internal labor; ensures rooms are ready faster through automated task logging. |

5. Technical Moat: Deep NLP and WhatsApp-First Interface

To ensure maximum adoption, ALTA utilizes a "No-App" interface. By leveraging WhatsApp—the primary communication channel in the region—we remove the barrier of app downloads and account creation.**Deep Arabic Dialect & NLP Requirements:**

- **Intent Extraction:** The system must go beyond keyword matching to understand the "Why" behind a guest's message.
 - **Linguistic Context & Dialects:** Native understanding of Saudi and Gulf dialects, including informal phrasing used in voice notes (Source Image 22).
 - **Sentiment Detection:** Automated detection of anger or urgency to trigger immediate human escalation.
- The Automated Request Journey (Voice-to-Action):**
1. **Input:** A guest sends a voice note in a local dialect: *"I need a room cleaning and a two-hour late check-out."*
 2. **Linguistic Processing:** ALTA's NLP extracts dual intents (Housekeeping + Booking Extension).
 3. **Autonomous Validation:** The system queries the PMS API to check room availability and payment status for the late check-out extension.
 4. **Execution & Confirmation:** ALTA logs the cleaning task for housekeeping and updates the PMS for the late check-out, sending an immediate confirmation to the guest.

6. Integration Architecture and Data Strategy

ALTA's architecture is designed to **"Unify Data in a Single Command Dashboard"** (Source Image 4). The platform acts as the "connective tissue" between existing hardware and software.

- **API Layer Approach:** ALTA connects to industry-standard PMS (Oracle, Mews) to pull real-time data on guest profiles, availability, and billing.
- **Phase 3 Predictive Intelligence:** The roadmap includes the development of occupancy forecasting and dynamic pricing engines. By analyzing accumulated data, the "Ultimate Brain" will suggest room price adjustments and personalized offers to **automatically increase profitability** (Source Image 54).
- **Data Unification:** ALTA provides management with a single source of truth, ending the era of cross-referencing multiple disjointed reports.

7. Financial Impact and ROI Framework

ALTA is a financial outcome, not a subscription expense. We sell the **Maximization of ROI**, which is the direct product of Operational Efficiency + Guest Experience (Source Image 41).**The Financial Impact Funnel:**

1. **Automated Inputs:** AI handles routine inquiries, reducing labor hours.
 2. **Operational Efficiency:** Faster response times and automated task closure lead to lower overhead.
 3. **Financial ROI:** Higher satisfaction scores drive positive Google Reviews, which increase direct booking confidence and reduce OTA commission dependency.
- The Impact Equation:** Reduction in Reception Calls + Faster Response Time + Increased Upselling = Maximized ROI

8. Implementation Roadmap: 90-Day MVP to 3-Year Vision

Our strategy prioritizes **"Validation Before Code,"** ensuring the AI models are trained on real-world Saudi hospitality data. | Phase | Timeline | Focus Area || ----- | ----- | ----- || **Phase 1: The Wedge (MVP)** | 90 Days | WhatsApp request management; intent classification; basic agent responses; manager dashboard. || **Phase 2: Expansion** | 1-2 Years | **Digital Reputation Agent** integration; marketing automation; guest recovery; full upsell automation. || **Phase 3: The Ultimate Brain** | 3 Years | Predictive AI (Dynamic pricing); **Multi-sector expansion (Hospitals & Residential Complexes)** ; Gulf regional expansion. |

Field Validation Plan:

1. **Discovery:** In-depth workflow mapping via interviews with **21+ industry experts** (Source Image 12).
2. **Semi-Manual Experiment:** A high-touch, **4-8 week validation study in a single hotel** to refine the NLP models and baseline the ROI (Source Image 26).
3. **ROI Measurement:** "Before vs. After" analysis of call volumes, response times, and guest sentiment.**Founding Vision:** We are not merely building a tool; we are establishing a **pioneering Saudi tech company** capable of regional and global expansion. ALTA AI is destined to be the "Integrated Brain" of the regional hospitality industry, scaling from 100 Saudi hotels to the broader Gulf market.